Lecture 5: Oligopoly

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Lecture 5 Outline

- Cournot model
  - Cournot model with n firms
  - Market power in Cournot competition
  - Cournot model with fixed costs
- Bertrand model
  - Bertrand model with capacity constraint
- Stackelberg model
  - Stackelberg model with n firms
- Cournot versus Stackelberg
- Extra Reading:
  - Church and Ware (2000) Industrial organization: a strategic approach Homewood, IL.: Irwin McGraw Hill. Chapter 8
  - Boyer and Moreaux (1987). Perfect competition as the limit of a hierarchical market game. Economics Letters.
  - Daughety (1990) Beneficial Concentration. The American Economic Review.



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Revision of Oligopoly models

- In oligopoly models we usually assume:
  - Consumers are price takers
  - There is a small number of firms
  - Firms produce a homogeneous product (one price)
  - Firms only compete in their prices or outputs (no other variables like advertising)
  - No possibility of new entrants (unless otherwise specified)

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Basic Cournot model

- Market demand: $p = a - bQ$, $Q = q_1 + q_2$. Constant costs.
- Firm 1's profit given q_2:
  $\pi_1 = p(q_1, q_2)q_1 - c_1 q_1 = (a - bq_1 - bq_2)q_1 - c_1 q_1$.
- FOD: $\frac{\partial \pi_1}{\partial q_1} = a - 2bq_1 - bq_2 - c_1 = 0$.
- Firm 1's reaction function: $q_1 = \frac{a - bq_2 - c_1}{2b}$.
- Similarly, firm 2's reaction function: $q_2 = \frac{a - bq_1 - c_2}{2b}$.
- Nash equilibrium: $q_1^* = \frac{a - 2c_1 + c_2}{3b}$, $q_2^* = \frac{a - 2c_2 + c_1}{3b}$.



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Basic Cournot model

- Firms set their output levels, aiming to maximise profit.
- Their revenue depends on their output and the market price, but the price depends on the total output: their own production plus the other's production.
- What happens if the firms decide simultaneously what to produce? This is Cournot's question.
- Essentially, Firm 1's decision depends on its EXPECTATION of, or its BELIEF about, how much Firm 2 will produce: How to correctly understand the Nash equilibrium in Cournot competition?

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Salient point

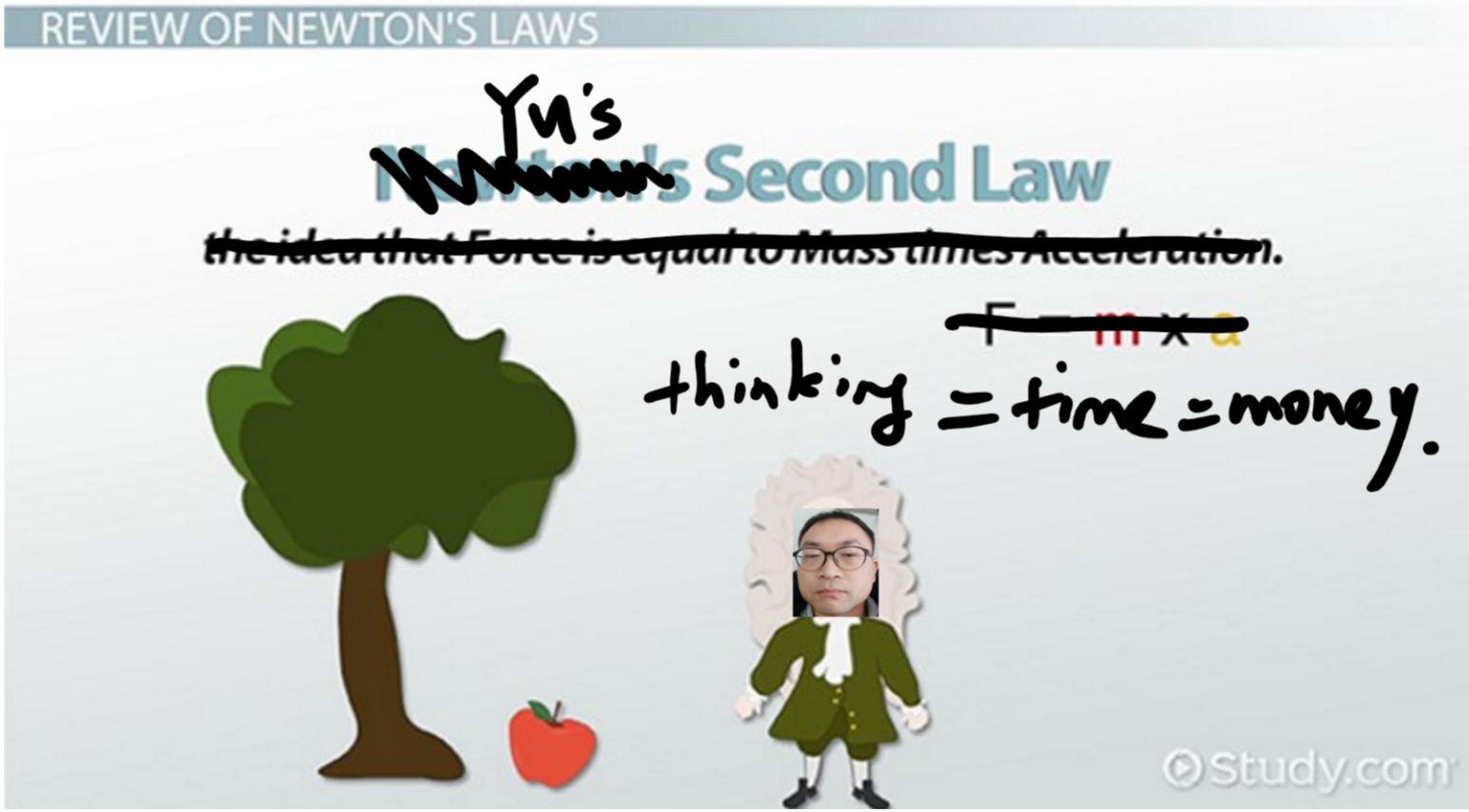
- Think of a number
- Think of a time
- Think of a location



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Basic Cournot model

- The value of the unique Nash equilibrium: Salient point.



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Extension 1: Cournot model with n firms

- Market demand: $p = a - bQ$, $Q = q_1 + q_2 + ...q_n$. Constant costs.
- q_{-i} refers to all other firms' provision except firm i.
- Firm i's profit given others:
  $\pi_i = p(q_i, q_{-i})q_i - c_i q_i = (a - bq_i - b\sum_{j \neq i} q_j)q_i - c_i q_i$.
- FOD: $\frac{\partial \pi_i}{\partial q_i} = a - 2bq_i - b\sum_{j \neq i} q_j - c_i = 0$.
- Firm i's reaction function: $q_i = \frac{a - b\sum_{j \neq i} q_j - c_i}{2b}$.

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Extension 1: Cournot model with n firms

- $$\begin{cases} a - c_1 = 2bq_1 + bq_2 + bq_3 ... + bq_n \\ a - c_2 = bq_1 + 2bq_2 + bq_3 ... + bq_n \\ a - c_3 = bq_1 + bq_2 + 2bq_3 ... + bq_n \\ ... \\ a - c_n = bq_1 + bq_2 + bq_3 ... + 2bq_n \end{cases}$$
- Adding up, $na - \sum_i c_i = (n+1)b\sum_i q_i$, so $b\sum_i q_i = \frac{na - \sum_i c_i}{n+1}$.
- Substitute into each $a - c_i$ equation: $a - c_i = bq_i + \frac{na - \sum_i c_i}{n+1}$, $q_i = \frac{a - c_i - \frac{na - \sum_i c_i}{n+1}}{b}$.
- When $c_i = c$ across the firms, $q_i = \frac{a-c}{(n+1)b}$. $Q = nq_i = \frac{n(a-c)}{(n+1)b}$.
- What if $n = 1$? Or n approaches infinity?



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Extension 2: market power in Cournot competition

- Given firm i's profit as: $\pi_i = p(q_i, q_{-i})q_i - c_i(q_i)$
- FOD: $\frac{\partial \pi_i}{\partial q_i} = p(q_i, q_{-i}) + \frac{\partial p(q_i, q_{-i})}{\partial q_i} q_i - MC_i(q_i)$.
- Since everyone is optimizing:
  $p(q_i^*, q_{-i}^*) + \frac{\partial p(q_i^*, q_{-i}^*)}{\partial q_i^*} q_i^* = MC_i(q_i^*)$.
- Re-arrange the equation, divide both sides by $p(q_i^*, q_{-i}^*)$ and manipulate things a bit:
  $\frac{p(q_i^*, q_{-i}^*) - MC_i(q_i^*)}{p(q_i^*, q_{-i}^*)} = -\frac{\partial p(q_i^*, q_{-i}^*)}{\partial q_i^*} q_i^* \frac{1}{p(q_i^*, q_{-i}^*)} \frac{Q^*}{Q^*}$



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Extension 2: market power in Cournot competition

Notice that

- The left side of the equation is the Lerner index L_i.
- Other variables remain unchanged, a change in q_i is equal to a change of Q: we can write $\frac{\partial p(q_i^*, q_{-i}^*)}{\partial q_i^*}$ as $\frac{\partial p(q_i^*, q_{-i}^*)}{\partial Q^*}$.
- The inverse of elasticity of market demand is $\frac{1}{\eta} = -\frac{\partial p(q_i^*, q_{-i}^*)}{\partial Q^*} \frac{Q^*}{p(q_i^*, q_{-i}^*)}$
- The equilibrium market share is denoted as $s_i = \frac{q_i^*}{Q^*}$
- In summary, we have $L_i = \frac{s_i}{\eta}$.

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Extension 2: market power in Cournot competition

From $L_i = \frac{s_i}{\eta}$, we have following observations:

1. The Cournot oligopolist will exercise market power. The equilibrium price exceeds the marginal cost of each firm ($s_i > 0$).
2. Cournot markups are less than monopoly markups ($s_i < 1$).
3. The market power of a Cournot oligopolist is limited by the market elasticity of demand. Greater η means less L.
4. There is an endogenous relationship between marginal cost and market share. Firms with lower marginal costs will have greater market shares: more efficient firms will be larger.
5. The higher the barriers to entry, the less number of competitors n, the greater each firm's market share s_i and the greater its market power L_i.



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Extension 3: Cournot model with fixed costs

- So far, in examining oligopoly, we have ignored fixed costs.
- But they are important for two reasons: 1. They may set the limit on the number of firms when there is free entry. 2. We need to consider them when we think about social welfare (market efficiency): on one hand, when more firms are competing, consumers benefit from a higher output and lower prices. On the other hand, if every firm has its own overheads, this can be a waste of resources.
- In other words, we need to address the over-competition issue, i.e., too many firms.



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Extension 3: Cournot model with fixed costs

- For example, take $p = 130 - 2Q$, $C_i(q_i) = 50 + 10q_i$
- Recall that $q_i = \frac{a-c}{(n+1)b} = \frac{130-10}{2(n+1)} = \frac{60}{n+1}$. Then...
- Market output $Q = nq_i = \frac{60n}{n+1}$
- Price $p = 130 - 2Q = \frac{130+10n}{n+1}$
- Firm revenue $pq_i = \frac{(130+10n)60}{(n+1)^2}$
- Firm cost $C_i = \frac{600}{n+1} + 50$
- Firm profit $pq_i - C_i = \frac{7200}{(n+1)^2} - 50$
- Total profit (produce surplus) $PS = \frac{7200n}{(n+1)^2} - 50n$
- Consumer surplus $CS = \frac{1}{2}(130 - \frac{130+10n}{n+1})\frac{60n}{n+1} = \frac{1800n^2}{(n+1)^2}$
- Total surplus $TS = CS + PS = \frac{1800n^2}{(n+1)^2} + \frac{7200n}{(n+1)^2} - 50n$



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Extension 3: Cournot model with fixed costs

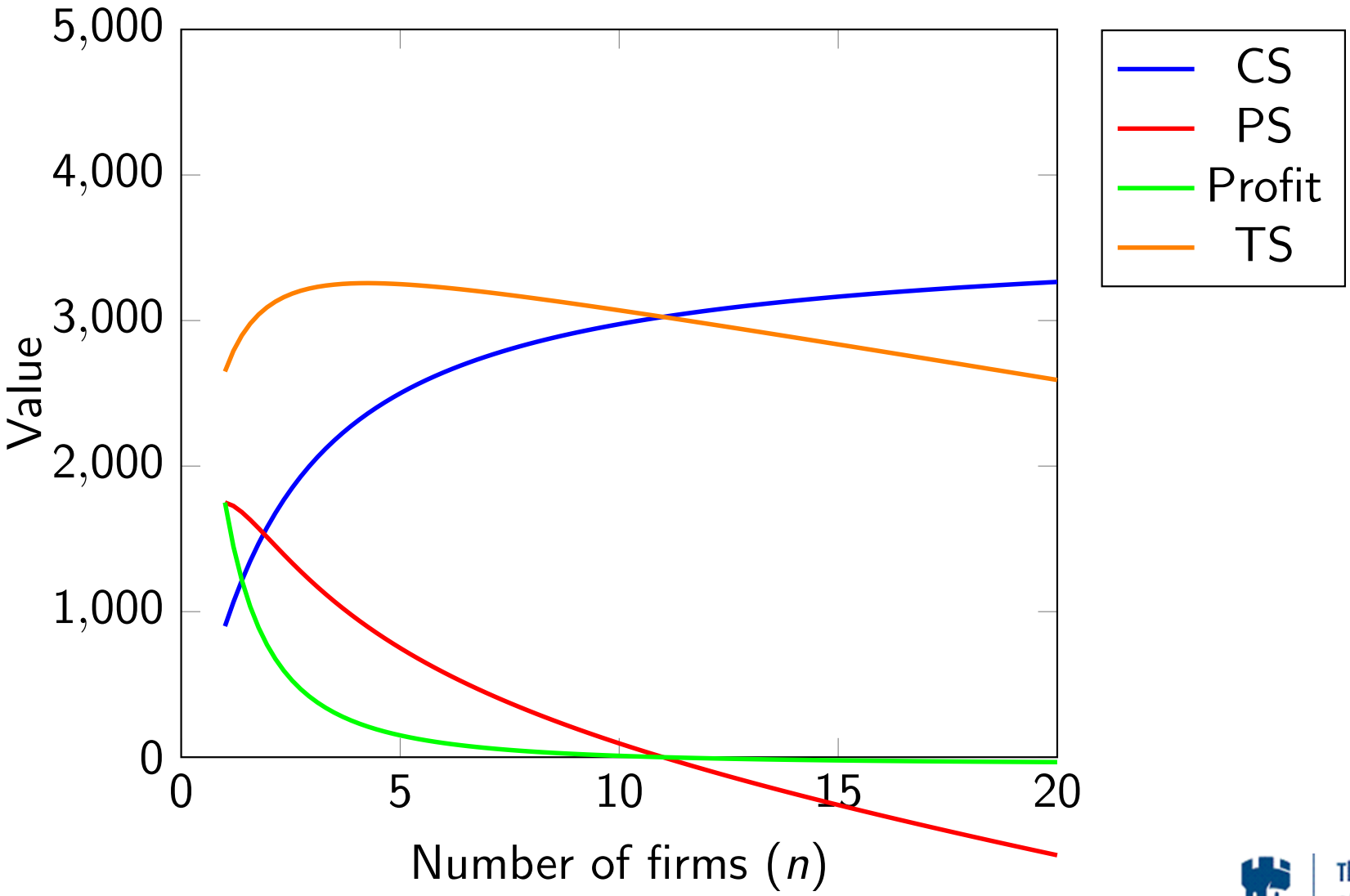
- Producer's welfare Matters!

No. of Firms	Cournot Firm Output	Market Output	Price	Firm Rev.	Firm Cost	Firm Profit	Total Profit	Cons. Surplus	Total Surplus	Fixed Cost 50
1	30.0	30.0	70.0	2100	350	1750	1750	900	2650	
2	20.0	40.0	50.0	1000	250	750	1500	1600	3100	
3	15.0	45.0	40.0	600	200	400	1200	2025	3225	
4	12.0	48.0	34.0	408	170	238	952	2304	3256	Social optimum
5	10.0	50.0	30.0	300	150	150	750	2500	3250	
6	8.6	51.4	27.1	233	136	97	581.6	2645	3227	
7	7.5	52.5	25.0	188	125	63	437.5	2756	3194	
8	6.7	53.3	23.3	156	117	39	311.1	2844	3156	
9	6.0	54.0	22.0	132	110	22	198	2916	3114	
10	5.5	54.5	20.9	114	105	10	95.04	2975	3070	
11	5.0	55.0	20.0	100	100	0	0	3025	3025	Equilibrium
12	4.6	55.4	19.2	89	96	-7	-88.8	3067	2979	



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Extension 3: Cournot model with fixed costs



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Basic Bertrand model

- In the Cournot and Stackelberg models, firms set quantities.
- This is realistic where firms have a long production lag and high sunk costs.
- Agricultural production is the usual example, and since agriculture was the main component of GDP in 19th Century France, it made sense.
- In modern service industries, or made-to-order manufactures, it makes more sense to advertise a product at a certain price, and then produce the quantity demanded at that price.
- That is what Bertrand's model assumes.
- That different assumption makes a huge difference to the outcome.



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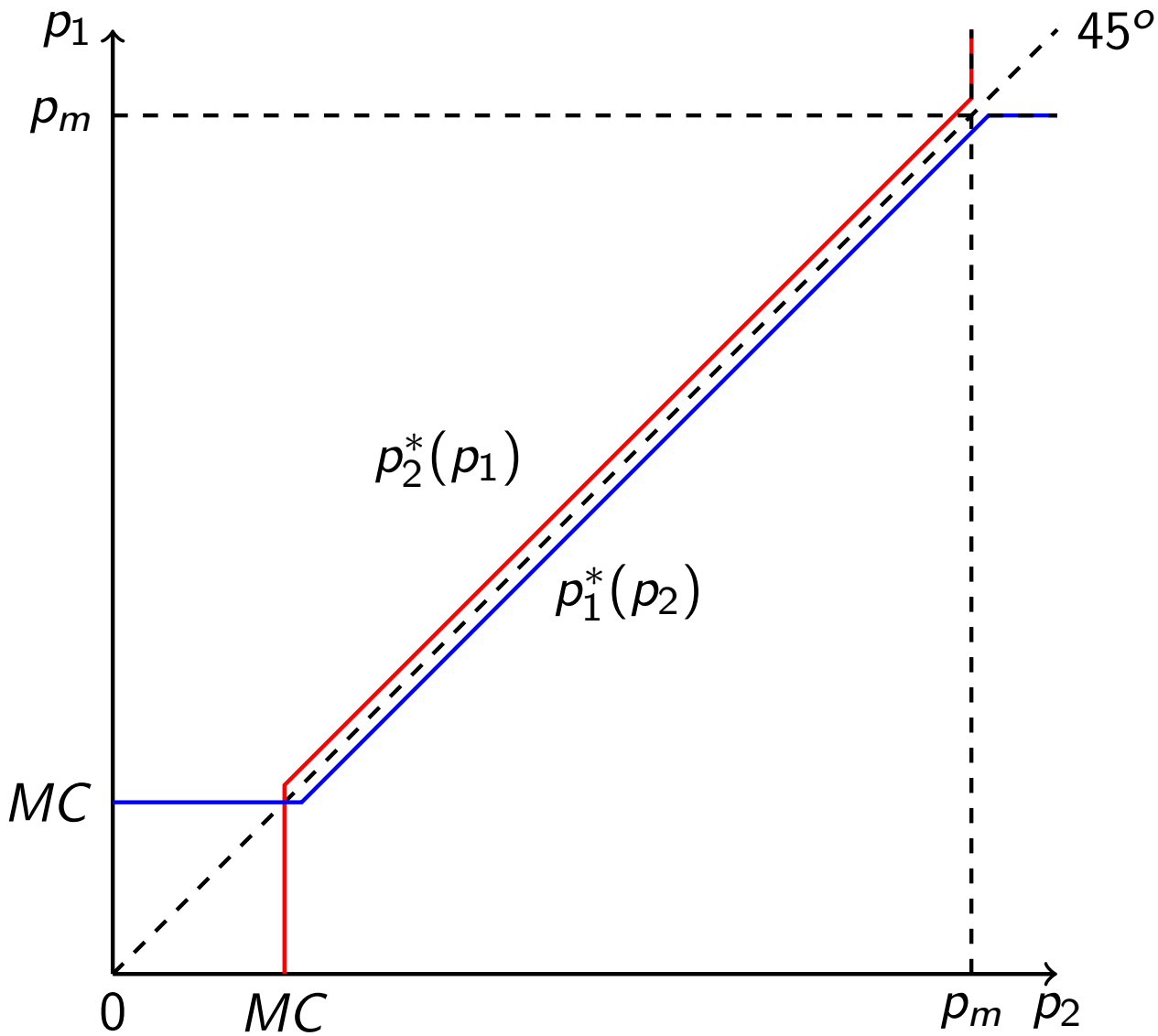
Basic Bertrand model

- If the firms sell identical goods, customers will always buy from the firms with the lower price.
- So, for any given p_2 that Firm 2 sets, Firm 1's best response is to set a price fractionally lower.
- The diagram shows the two reaction functions.
- p_1 as a function of p_2: $p_1 = p_2 - \epsilon$, $\epsilon > 0$ and ϵ approaches 0. In the graph, it is just below the 45 degree line. Firms 2's reaction function is symmetrical.
- But neither firm will set a price below marginal cost. So the reaction functions meet at $p_1 = p_2 = MC$. That is a Nash Equilibrium.
- What if two firms have different MC?
- Where does the upper limit price p_m come from?



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Basic Bertrand model



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Basic Bertrand model

- Thus the model has the same outcome as pure competition.
- Why is this model interesting?
- In this form, it is interesting because it shows how important the quantity-setting assumption is, to achieve non-competitive results in the other models.
- It becomes more interesting when we consider:
  - The effect of capacity constraints
  - Oligopoly models with heterogeneous goods (discussed in the future chapter)

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Extension 1: Bertrand model with capacity constraint

- In the Bertrand duopoly model, firms are price setters. Each firm's best response is to undercut the other firm's expected price and gain the entire market.
- The only price at which neither firm has an incentive to raise or lower its price is at the marginal cost, where the two firms presumably share the market equally and both earn zero profit — assuming that either firm can satisfy the market demand easily.
- But what if both firms have a capacity constraint? For example, when the market demand is $p = 1 - Q$, for each firm, $MC = 0.2$ when $q \leq 0.4$ and their MC curves are vertical when $q > 0.4$.
- It seems like the firms' total production can satisfy the market demand. But is $q_1 = q_2 = 0.4$, $p = MC = 0.2$ a Nash equilibrium?



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Extension 1: Bertrand model with capacity constraint

- Given $p_1 = p_2 = 0.2$, market demands is 0.8. When firm 1 provides $q_1 = 0.4$, firm 2 has an incentive to deviate: if it produces less than 0.4, then the market supply is less than the market demand, there are frustrated customers who are willing to pay higher prices. Therefore, firm 2 is now effectively a monopolist with respect to this market, with its own residual demand and a corresponding marginal revenue schedule as shown by the blue curve in the graph. It can now maximise profit by charging $p_2 = 0.4$ and produce $q_2 = 0.2$.
- Then, firm 1 has an incentive to increase its profit by charging higher prices: as long as it charges $p_1 < p_2 = 0.4$, it can sell at a higher price without selling less. Therefore, firm 1 charges a price equal to $p_1 = 0.4 - \epsilon$ instead.



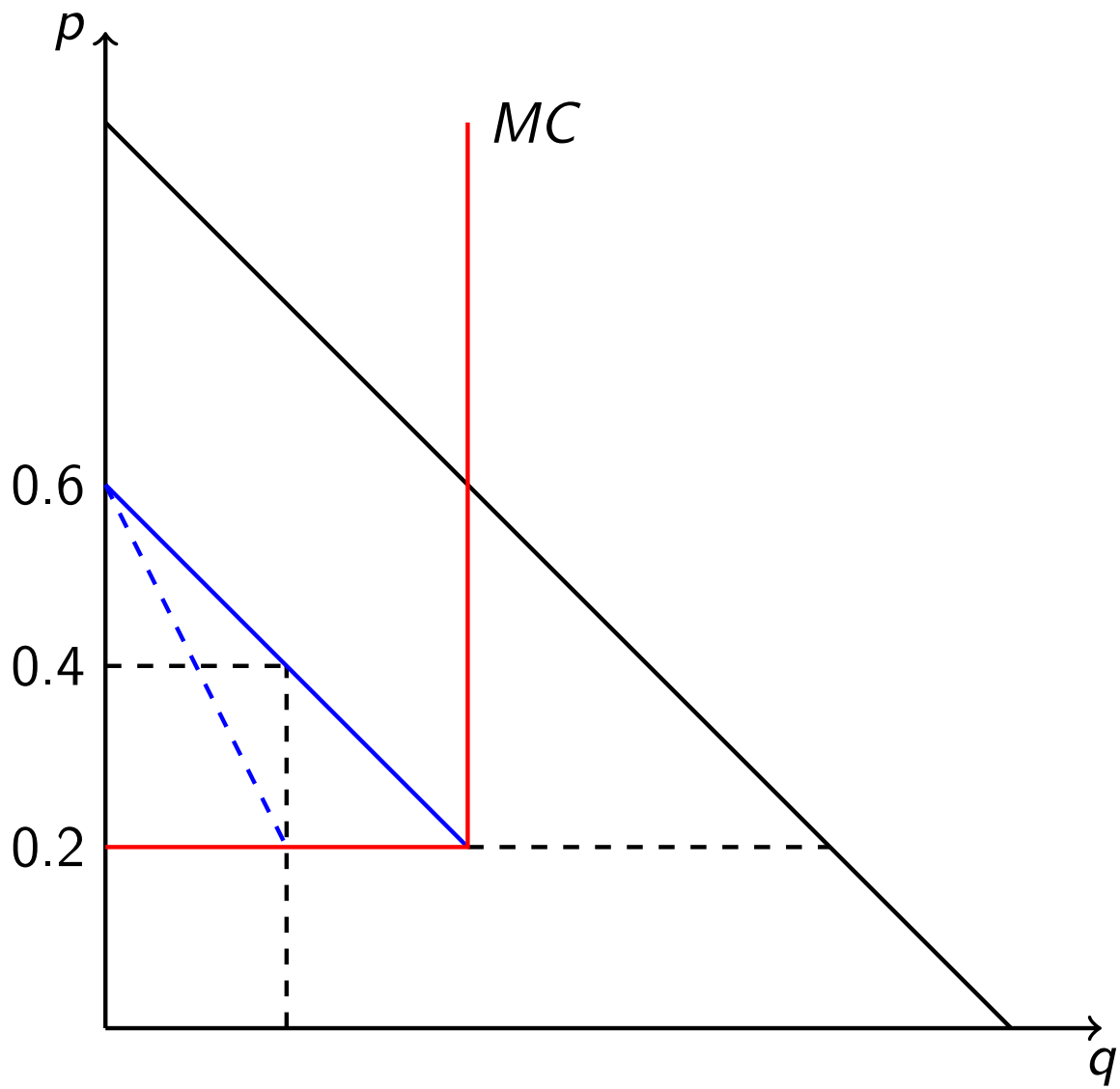
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Extension 1: Bertrand model with capacity constraint

- But Firm 2 would respond by charging $p_2 = 0.4 - 2\epsilon$ now: by doing so, it can double its selling (from 0.2 to 0.4) without sacrificing too much in the selling price.
- This dance continues, each player wishes to charge lower price to maximize its producing capacity, until $p_1 = p_2 = 0.2$, both firms reach their maximum capacity. But then, it goes around one more time.
- Now we have proven that $q_1 = q_2 = 0.4$, $p = MC = 0.2$ is not a Nash equilibrium. Actually, there is no pure strategy Nash equilibrium.
- Question: Is there a Nash equilibrium though?

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Extension 1: Bertrand model with capacity constraint



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Extension 1: Bertrand model with capacity constraint (optional)

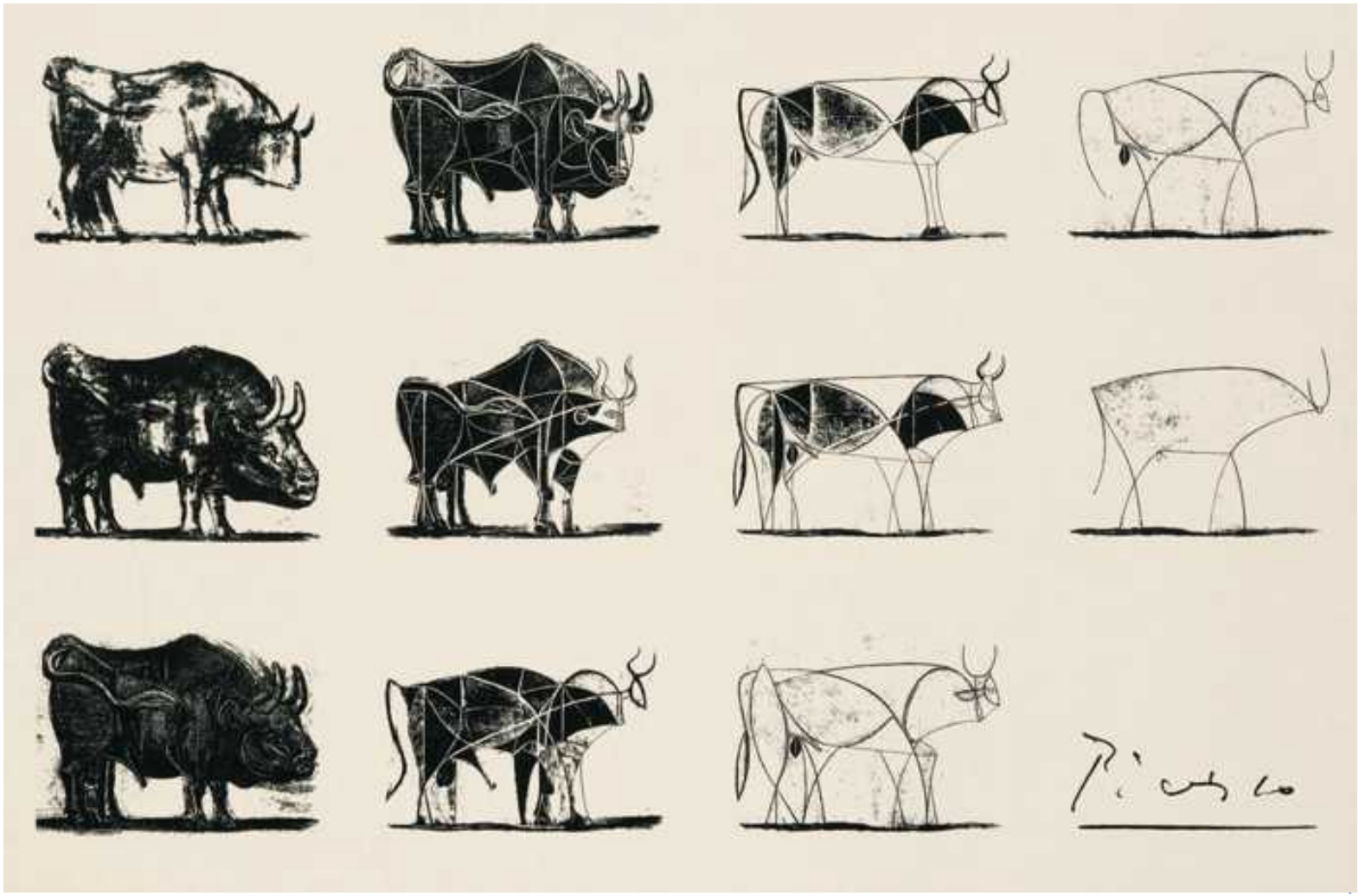
- A slightly more formal proof for the non-existence of pure strategy Nash equilibrium:
- 1. Suppose that $p_1 > p_2 > 0.2$, then firm 2 sells 0.4 and would increase the price to $p_1 - \epsilon$. Same goes for $p_2 > p_1 > 0.2$
- 2. Suppose that $p_1 = p_2 > 0.2$, then each firm has an incentive to slightly lower the price to sell 0.4.
- 3. Suppose that $p_1 = p_2 = 0.2$, then each firm has an incentive to charge the residual monopoly price instead (as shown before).
- Therefore, all suppositions fail and there is no pure-strategy equilibrium: the price is indeterminate.



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Now think:

What is the essential difference between Cournot competition and Bertrand competition?



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Basic Stackelberg model

- The Cournot model assumes that firms have to decide their outputs simultaneously.
- But what if one firm moves ahead of the other one? That was Stackelberg's idea.
- If one firm (the leader) can commit itself to a certain output level, then it knows exactly what the other firm (the follower) will do.
- It can base its own output choice on this knowledge.



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Basic Stackelberg model

- Market demand: $p = a - bQ$, $Q = q_1 + q_2$. Constant costs. Firm 1 leads and firm 2 follows.
- Firm 2's profit after observing q_1:
  $\pi_2 = p(q_1, q_2)q_2 - c_2 q_2 = (a - bq_1 - bq_2)q_2 - c_2 q_2$.
- FOD: $\frac{\partial \pi_2}{\partial q_2} = a - bq_1 - 2bq_2 - c_2 = 0$.
- Firm 2's reaction function: $q_2 = \frac{a - bq_1 - c_2}{2b}$.



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Basic Stackelberg model

- Anticipating 2's reaction function, firm 1's profit:
  $\pi_1 = p(q_1, q_2(q_1))q_1 - c_1 q_1 = (a - bq_1 - bq_2(q_1))q_1 - c_1 q_1 = (a - bq_1 - b\frac{a-bq_1-c_2}{2b})q_1 - c_1 q_1 = \frac{a-bq_1+c_2}{2}q_1 - c_1 q_1$.
- FOD: $\frac{\partial \pi_1}{\partial q_1} = \frac{a-2bq_1+c_2}{2} - c_1 = 0$.
- Firm 1's equilibrium strategy: $q_1^* = \frac{a-2c_1+c_2}{2b}$.
- Firm 2's equilibrium strategy: $q_2^* = \frac{a-bq_1^*-c_2}{2b} = \frac{a+2c_1-3c_2}{4b}$.



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Extension 1: Stackelberg model with n firms (Boyer and Moreaux 1987)

- Market demand: $p = a - bQ$, $Q = q_1 + q_2 + ...q_n$. Same constant costs $c_i = c$ across firms. Firms make their production in the sequence of $1, 2, 3...n$. Denote $Q_i = \sum_{j=1}^{i-1} q_j$ as the ACCUMULATED production BEFORE (not including) firm i.
- Firm n's profit $\pi_n = p(q_n, Q_n)q_n - cq_n = (a - c - bq_n - bQ_n)q_n$.
- FOD: $\frac{\partial \pi_n}{\partial q_n} = a - c - 2bq_n - bQ_n = 0$.
- Firm n's reaction function: $q_n = \frac{a-c-bQ_n}{2b}$.

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Extension 1: Stackelberg model with n firms

- Firm $n-1$'s profit

$$\pi_{n-1} = p(q_{n-1}, Q_{n-1}, q_n)q_{n-1} - cq_{n-1}$$

$$= (a - c - bq_{n-1} - bQ_{n-1} - b\frac{a - c - bQ_n}{2b})q_{n-1}$$

$$= (a - c - bq_{n-1} - bQ_{n-1} - b\frac{a - c - b(q_{n-1} + Q_{n-1})}{2b})q_{n-1}$$

$$= \frac{a - c - b(q_{n-1} + Q_{n-1})}{2}q_{n-1}$$

- FOD: $\frac{\partial \pi_{n-1}}{\partial q_{n-1}} = a - c - 2bq_{n-1} - bQ_{n-1} = 0.$
- Firm $n-1$'s reaction function: $q_{n-1} = \frac{a-c-bQ_{n-1}}{2b}$.

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Extension 1: Stackelberg model with n firms

- From $q_n = \frac{a-c-bQ_n}{2b}$, $q_{n-1} = \frac{a-c-bQ_{n-1}}{2b}$... we can reasonably guess that $q_i = \frac{a-c-bQ_i}{2b}$ for every i. (The rigorous proof is not too difficult for you to try.)
- At $i = 1$, $Q_1 = 0$. $q_1 = \frac{a-c-bQ_1}{2b} = \frac{a-c}{2b}$.
- At $i = 2$, $Q_2 = q_1$. $q_2 = \frac{a-c-bQ_2}{2b} = \frac{a-c-b\frac{a-c}{2b}}{2b} = \frac{a-c}{4b}$.
- At $i = 3$, $Q_3 = q_1 + q_2$.
  $q_3 = \frac{a-c-bQ_2}{2b} = \frac{a-c-b\frac{a-c}{2b}-b\frac{a-c}{4b}}{2b} = \frac{a-c}{8b}$.
- This process goes on, in equilibrium, $q_i = \frac{a-c}{2^i b}$,
  $Q = \sum_{i=1}^{n} q_i = (\frac{1}{2} + \frac{1}{4} + \cdots \frac{1}{2^n})\frac{a-c}{b} = (1 - \frac{1}{2^n})\frac{a-c}{b}$.



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Cournot versus Stackelberg I

- Compare the equilibrium output in Cournot with n firms and Stackelberg with n firms. Which case has a higher total output? A lower price? What about the consumer/producer/total welfare?
- WHY?

	$n=2$	$n=3$	$n=4$
Cournot	$(\frac{1}{3}, \frac{1}{3})$	$(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$	$(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5})$
Stackelberg	$(\frac{1}{2}, \frac{1}{4})$	$(\frac{1}{2}, \frac{1}{4}, \frac{1}{8})$	$(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16})$



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Cournot versus Stackelberg II (Daughety 1990)

- What about a combination of Cournot and Stackelberg model?
- Suppose that in a two-periods game, we have m identical firms competing with each other and n identical firms competing with each other. Market inverse demand function is $p = a - bQ$ as usual. Each firm has a cost c.
- In the same period, firms are competing in the Cournot style, while across the periods, firms are competing in the Stackelberg style.
- Suppose that each leader firm produces q_l, in total, $Q_l = mq_l$. Similarly, each follower firm produces q_f, in total $Q_f = nq_f$.



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Cournot versus Stackelberg II

- In the second period, the market inverse demand function $p = a - b(Q_l + Q_f)$ can be written as $p = (a - bQ_l) - bQ_f$, taking Q_l as given. From last lecture's result in Cournot model extension 1, it can be easily derived that $q_f = \frac{a-bQ_l-c}{(n+1)b}$, $Q_f = n\frac{a-bQ_l-c}{(n+1)b}$.
- In the first period, anticipating Q_f, the market inverse demand function is $p = a - b(Q_l + Q_f) = (a - bQ_l - bn\frac{a-bQ_l-c}{(n+1)b}) = \frac{a+nc}{n+1} - \frac{b}{n+1}Q_l$. Similarly, it can be easily derived that $q_l = \frac{\frac{a+nc}{n+1} - c}{(m+1)\frac{b}{n+1}} = \frac{a-c}{(m+1)b}$, $Q_l = \frac{m(a-c)}{(m+1)b}$.
- Accordingly, $q_f = \frac{a-c}{(m+1)(n+1)b}$, $Q_f = \frac{n(a-c)}{(m+1)(n+1)b}$.
- Have a try: $m = 1$ and $n = 1$, $m = 2$ and $n = 0$, $m = 0$ and $n = 2$...



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Cournot versus Stackelberg II

- Then what?
- Same old, same old. Higher total output? Lower price? Consumer/producer/total welfare? ...
- Pros and cons for a merger?



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Cournot versus Stackelberg II

- Feltovich (2001): (Voluntary) Merger in Daughety's model
- Huck, Konrad and Muller (2001): Only merger profitable for firms in Daughety's model is the merger between a leader and a follower.
- Escrihuela-Villa and Fauli-Oller (2007): Merger when follower are less efficient than leader in Daughety's model
- Escrihuela-Villa and Fauli-Oller (2019): Convex costs version of Escrihuela-Villa and Fauli-Oller (2007)
- Scrimitore (2013): Product differentiation (substitutability) in Daughety's model
- Yoshida (2016): Product differentiation, Leaders have different costs while followers have the same costs



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Thank you



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